

# CODEMIX SKILL

The safety + decision layer between AI agents and their actions

**We put an intent-validation safety gate between what the voice agent hears and what the agent is allowed to do.**

Initiative: Codemix Skill • Owner: Team Ramanathan & Sadhana • Track 1: Customer & Employee Experience

## ● THE GAP

Voice agents can immediately call a refund, cancellation or replacement API. One missed negation turns the right conversation into the wrong transaction.

## ● OUR PRODUCT

A plug-in between the AI agent and its API/tool calls. It validates intent, negation, confidence, sentiment and SOP rules before execution.

## ● FRESHWORKS EXTENSION

Unsafe, uncertain or escalated calls become Freshdesk tickets with transcript, decision context and action details for a human agent.

Understand

Decide

Act

Audit

*“Order cancel **mat** karo, bas address change karna hai.”*

## 2

# Problem Statement

*In code-mixed voice, one short word can reverse the meaning of an API action*

## HINDI EXAMPLE

*“Order cancel **mat** karo, bas address change karna hai.”*

**Correct intent: do NOT cancel → update address**

## TAMIL EXAMPLE

*“Refund **vendam**, replacement anuppunga.”*

**Correct intent: do NOT refund → send replacement**

## HOW IT BREAKS TODAY

Caller says “don't cancel”

Speech-to-text drops  
“mat”, or the intent model  
ignores it

Agent calls cancel\_order()

**Order cancelled: money,  
reshipping, trust lost**

**48%**

of negated requests would still trigger a  
refund / cancel in our own Round-1  
engine

*Measured: 11 of 23 team-written test sentences  
(Hindi, Tamil, English, Telugu, Kannada, Bengali)*

**26%**

of cash-on-delivery orders in India return  
to origin, so refund / cancel / return calls  
are high-volume

*Source: Shipway ShipNotes report, FY25 (Jul 2025)*

# Why Now & Competitive Landscape

*AI agents have write access, and nobody gates what they are allowed to do on a misheard word*

## WHY NOW



### AI agents can act

Tool calling and MCP let agents trigger refunds, cancellations and returns directly.



### Code-mixing is normal

Indian callers switch language mid-sentence; the risk sits in small intent-flipping words.



### The voice stack is ready

VoBiz, Sarvam, Claude and ElevenLabs make a checked, real-time voice agent possible now.

**27-70%** **Hinglish word error rate across speech-to-text models.** Speech recognition alone can't be trusted with irreversible actions. *Deepgram*

**Our timing** Not another voice bot: the layer that decides when an agent may call an API.

## COMPETITIVE LANDSCAPE

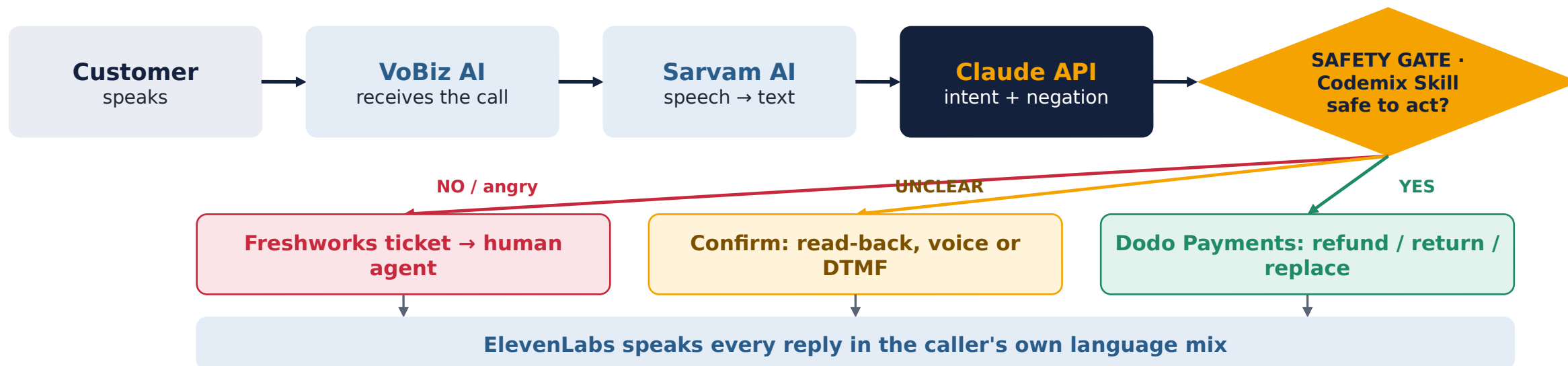
| Alternative            | Strength                                                 | Gap                                                            |
|------------------------|----------------------------------------------------------|----------------------------------------------------------------|
| Do nothing             | Zero effort                                              | Risky actions run on the first interpretation                  |
| Confirm-everything IVR | Safe confirmation                                        | Friction on every action, even low-risk ones                   |
| Generic LLM guardrails | Policy, toxicity, PII                                    | Not an action gate for code-mixed negation + speech confidence |
| Codemix Skill          | Code-mixed intent → risk check → ALLOW / CONFIRM / BLOCK | Unsafe cases go to Freshworks with full context                |

**Our edge** Risk-based action gate + human escalation + a full audit record.

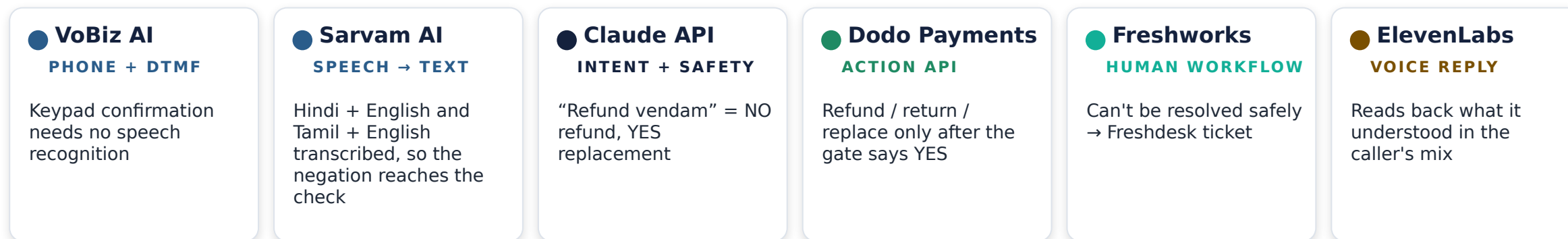
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# Solution: the Intent-Validation Safety Gate

*Between what the voice agent hears and what it is allowed to do; each part has one job in preventing a wrong action*

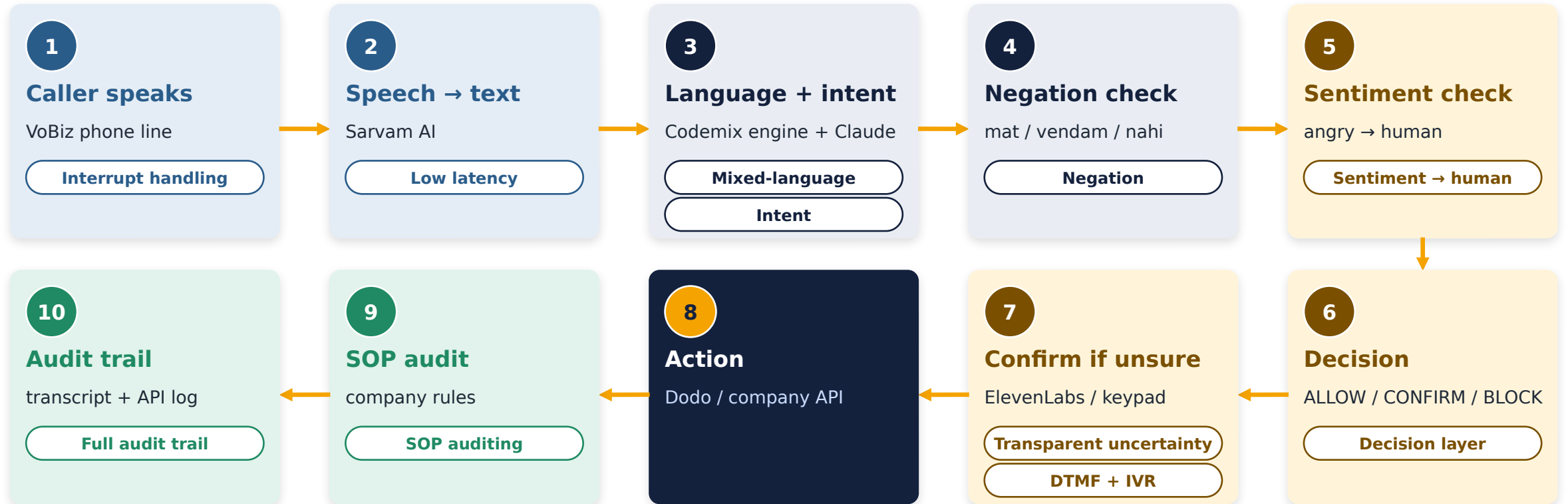


## HOW EACH PART PREVENTS A WRONG ACTION



# 5 One Call, End to End — All 11 Features

*Every feature appears only where it adds value; routine calls take the fast path*



**Sentiment:** Claude scores caller sentiment every turn; two negative turns or an explicit request for a human → Freshdesk ticket.

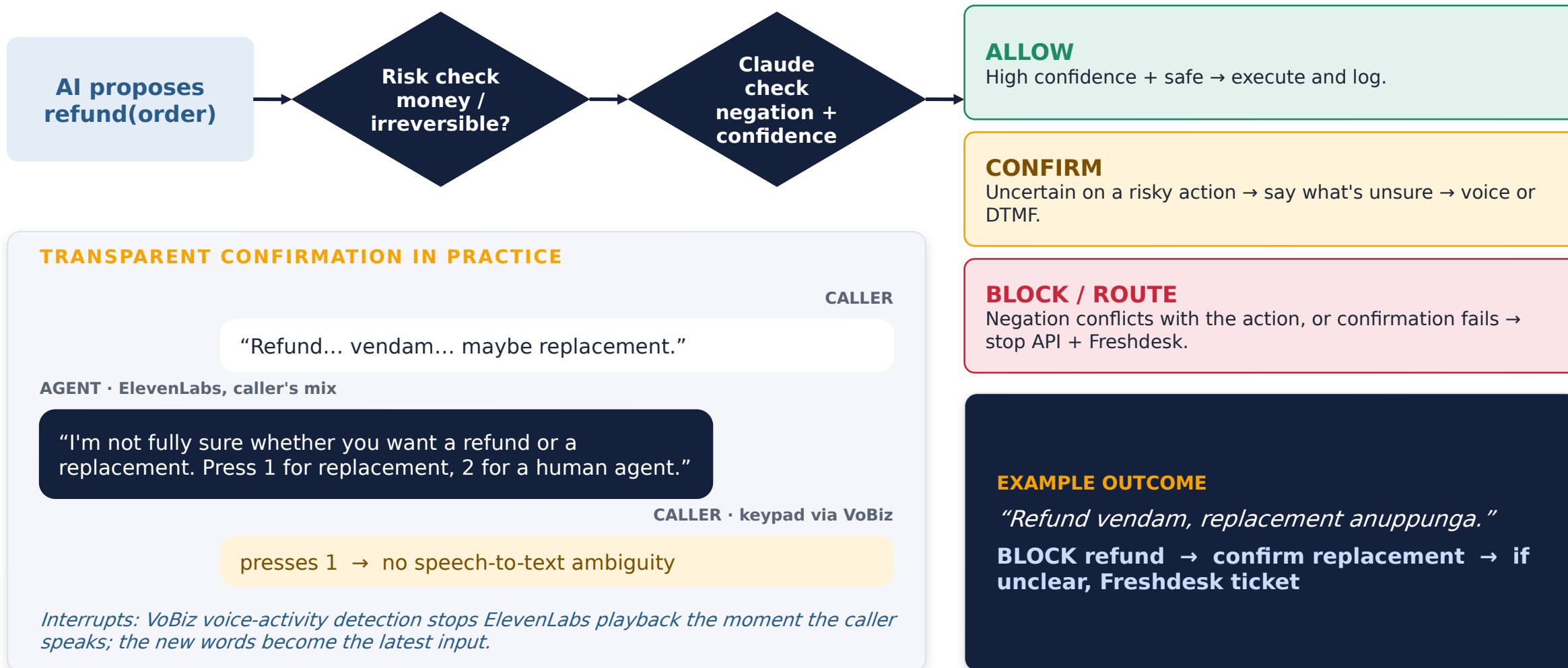
**Low latency:** fast offline checks first (1.87 ms measured); Claude runs only before sensitive actions.

**SOP audit:** every call is checked against company rules (authorisation, required confirmation, escalation) → PASS / VIOLATION + audit reference; transcript, decisions and API calls are stored.

## 5

# The Safety Decision & Transparent Confirmation

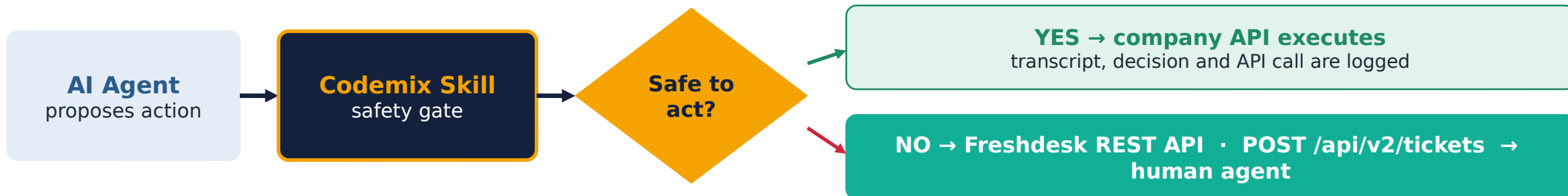
*The agent never silently guesses on an irreversible action*



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# Freshworks Integration — Human-in-the-Loop

*Freshdesk takes the handoff; also works with Freshcaller as the voice channel and alongside Freddy AI Agent*



## A ticket is created when...

- the action is unsafe or unclear
- the caller doesn't confirm
- the caller is angry / distressed
- the AI should not guess



## What we add to Freshdesk

Code-mixed intent validation, action gating, transparent confirmation and structured escalation, on top of ticketing, agent workspace and customer history.

## What the human agent receives in the Freshdesk ticket

### CONVERSATION

Call ID + timestamp

Original code-mixed transcript

Detected languages

English summary

Voice / DTMF source

### DECISION

Intent + confidence

Negation detected

Safety verdict

Requested API action

Confirmation status

### OPERATIONAL

Priority / tags

SOP result

Reason for escalation

Attempted API call

Audit reference

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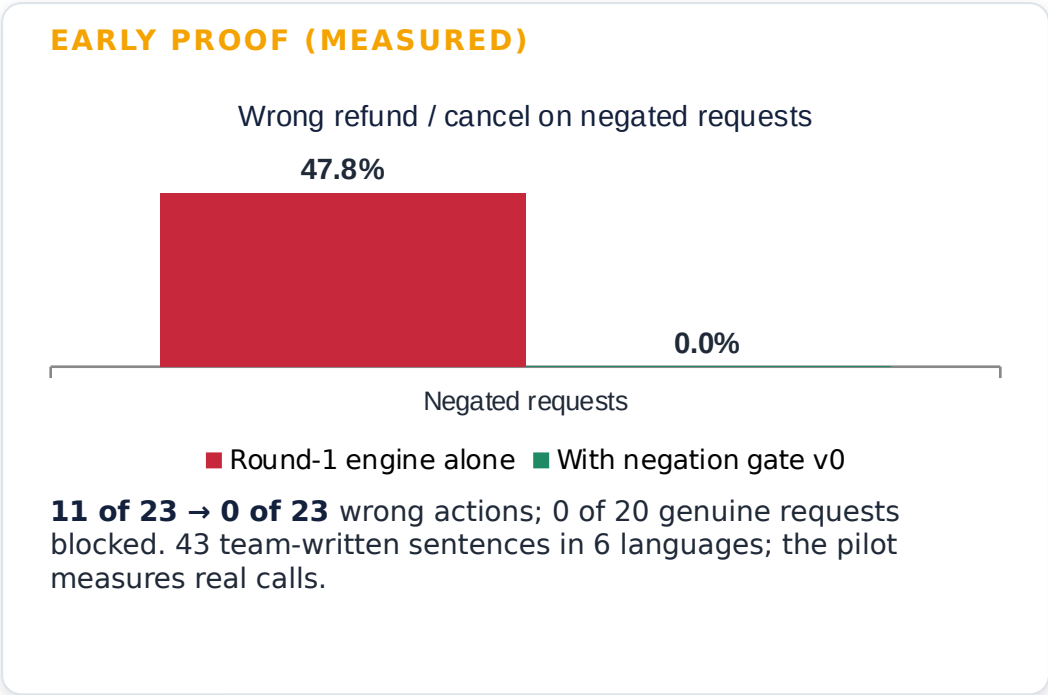
# Agentic Design, MVP Scope & Early Proof

*A plug-in control point that intercepts the proposed action before the company API is called*

AGENTIC DESIGN: THE PLUG-IN CONTROL POINT



|                |                                                                                                                                               |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| BUILT TODAY    | Code-mix engine + intent scoring · MCP server · live Freshdesk ticket API · negation gate v0 + benchmark · Vercel demo                        |
| MVP (3 MONTHS) | VoBiz + Sarvam + ElevenLabs voice path · Claude safety gate · Dodo action path · Freshdesk escalation · Hindi + Tamil + English · audit trail |
| NEXT           | Sentiment + interrupt tuning · SOP rule editor + reports · more Indian languages · Freshworks Marketplace app                                 |



# Customers, Value, Pricing & Go-to-Market

*Start where AI agents already have permission to act; pilot → proof → Freshworks distribution*

## 6 Target Customers

**Primary:** D2C, e-commerce and fintech brands on Freshdesk deploying AI voice agents for Indian customers.

**Secondary:** BPOs needing one safety / policy layer across many clients.

**Protected actions:** refunds, returns, replacements, cancellations.

*Why now: ~26% of COD orders return to origin (Shipway FY25), so these calls are high-volume.*

## 7 Value Proposition

**Plug it in once: the existing agent serves mixed-language callers, and nothing irreversible happens on a misheard word.**

### Without

- Acts on first guess
- Dropped “mat” = wrong refund
- Blind escalation, no record

### With Codemix Skill

- Checks intent + negation first
- Confirms when unsure
- Freshdesk handoff + audit trail

## 8 Pricing & Packaging (proposed)

### Free

Monitor mode: flags risky actions, never blocks

### \$0.03

per guarded action: plug-in on the customer's own agent

### ₹8 / min

full voice agent: VoBiz + Sarvam + Claude + ElevenLabs + Freshdesk

### Custom

Enterprise: SOP rule packs, India data residency, SLA

*Priced above run cost (~55% gross margin, slide 10). One prevented ₹1,000 wrong refund pays for ~390 guarded actions.*

## 9 Go-to-Market Plan

- 1 3 Indian D2C design partners on Freshdesk
- 2 Free monitor-mode pilot
- 3 Publish wrong-action proof + demo video
- 4 Freshworks Marketplace listing
- 5 Scale via Freshdesk deals + BPO partners

# Costs, Savings & Revenue, Metrics & Risks

*What it costs, who gains what, how we measure it, and what could go wrong*

## 10 Costs & Unit Economics

~\$90K

3-month MVP build: M1 data · M2 gate + voice · M3 Freshdesk + pilot

Engineers \$60K · linguist \$10K · PM \$8K · APIs \$7K · buffer \$5K

### RUN COST PER 3-MIN VOICE CALL

|                                      |              |
|--------------------------------------|--------------|
| VoBiz telephony (₹0.45/min)          | ₹1.35        |
| Sarvam STT (₹30/hr)                  | ₹1.50        |
| ElevenLabs Flash (\$0.05/1K chars)   | ₹6.40        |
| Claude checks (Haiku 4.5 / Sonnet 5) | ₹0.71        |
| Hosting + transcript storage         | ₹0.43        |
| <b>Total ≈ ₹3.5/min vs ₹8 price</b>  | <b>₹10.4</b> |

*Plug-in only: ₹1.1 cost vs ₹2.55 price per action. Dodo and Freshdesk fees stay on the customer's own plans.*

## 11 Success Metrics

|                           | TODAY   | TARGET |
|---------------------------|---------|--------|
| Wrong actions on negation | 48%*    | <0.5%  |
| Engine latency            | 1.87ms† | <5ms   |
| p95 reply (sensitive)     | —       | <900ms |
| Confirmation rate         | —       | ≤15%   |
| Audit coverage            | —       | 100%   |
| Reversal tickets          | —       | −30%   |
| Adoption (2 qtrs)         | —       | 15%    |

*\* Round-1 engine, no gate, 23-sentence test set † npm test average*

## 12 Risks & Assumptions

**STT drops the negation**

→ confirm every risky action

**Safety check adds latency**

→ deep checks only when sensitive

**Dialect / spelling variation**

→ lexicon + model fallback

**Read-back friction**

→ ≤15% confirmation budget

### ASSUMPTIONS

- Pilot brands expose refund / return APIs
- Callers accept a short confirmation
- Sarvam is accurate enough with the fallback

## WHO GAINS WHAT IN YEAR 1

*illustrative · 48% misfire rate measured, volumes assumed · details in speaker notes*

### BRANDS SAVE

≈ ₹6 Cr

wrong refunds & cancellations avoided · ₹3.8 saved for every ₹1 paid

### FRESHWORKS EARNS

\$349K

Plug-in \$180K + Voice agent \$169K

### GROSS PROFIT

~\$195K

after \$154K run costs · pays back the \$90K build in ~8 months